6	**(a)**	A magnetic field is a region of space in which a magnetic force is experienced by **moving** charges or current-carrying conductors or permanent magnets placed in this field.	A1
	(b)(i)	As the positive ion passes into the region of magnetic field, by Flemming's left hand rule, there is a magnetic force on the ion towards the positive plate. Due to the electric field, there will be a electric force on the ion towards the negative plate. If the magnitude of these two forces are equal, the ion will be able to pass through without deviation. (If student state "Electric force is opposite in direction to magnetic force" award only 1 mark out of the first 2 B1 marks).	B1 B1 B1
	(b)(ii)1.	In the region with B-field only there exist a magnetic force (of constant magnitude) perpendicular to direction of motion causes ions to undergo centripetal acceleration / provides the centripetal force.	A1
	(b)(ii)2.	$B'qv = \frac{mv^2}{r}$ $\frac{m}{r} = \frac{B'q}{v}$ = constant $\frac{M}{26.2} = \frac{12u}{22.4}$ $M = 14u$	C1 A1

7	**(a)(i)**	Photoelectric effect is the ejection/emission of an electron from a metal surface	A1